Term 3 — Lesson 1 Practice Activity 3 (C1–C3) with Solutions

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Evaluated criteria

- C1: Identify whether the context uses a population proportion or a population mean and map the symbols correctly in context.
- C2: Write the null and alternative hypotheses in correct notation, keeping the null as “parameter equals benchmark.”
- C3: Classify the test direction (left-tailed, right-tailed, or two-tailed) and justify it from the wording of the claim.

1 Problem 1 — Reusable bottle participation in Grade 11

The student council wants to know whether the true proportion of Grade 11 students who bring a reusable water bottle is now above the campaign benchmark of 60%. During one morning check, they record a sample of $n = 150$ students and observe $x = 99$ students with reusable bottles.

Questions/Tasks.

(C1) Identify whether this is a proportion or mean setting, define the parameter, and map the given sample information.

(C2) Write the null and alternative hypotheses for the claim.

(C3) Classify the tail type and justify it from the context.

C1

This is a population proportion setting. Let p be the true proportion of Grade 11 students who bring a reusable bottle on a regular school day. The benchmark is $p_0 = 0.60$. The sample information is $n = 150$, $x = 99$, and $\hat{p} = 99/150 = 0.66$.

C2

$$H_0 : p = 0.60, \qquad H_a : p > 0.60.$$

In words, the null keeps the historical benchmark at 60%, and the alternative represents the campaign claim that the proportion is higher than 60%.

C3

This is a right-tailed test because the claim is "above 60%." The alternative uses $>$, so evidence is looked for in the upper tail.

2 Problem 2 — On-time first submission rate in economics class

The teacher tracks whether students submit the first weekly task on time. The department benchmark says the true on-time proportion should be at least 85%, and the teacher worries that the real rate may have dropped below that level. In a sample of $n = 120$ students, $x = 94$ submitted on time.

Questions/Tasks.

(C1) Identify the parameter type and map the symbols in context.

(C2) Write the null and alternative hypotheses.

(C3) Classify and justify the tail type.

C1

This is a population proportion setting. Let p be the true proportion of students who submit the first weekly task on time. The benchmark is $p_0 = 0.85$. The sample summary is $n = 120$, $x = 94$, and $\hat{p} = 94/120 \approx 0.783$.

C2

$$H_0 : p = 0.85, \qquad H_a : p < 0.85.$$

The null represents the benchmark level, and the alternative captures the concern that punctual submission is lower than the target.

C3

This is a left-tailed test because the question is whether the true on-time proportion is below the benchmark. The alternative uses $<$.

3 Problem 3 — Average waiting time in the school cafeteria

After changing lunch-line organization, school leaders want to verify whether the true mean waiting time is now less than 9 minutes. A time study of $n = 64$ students reports a sample mean of $\bar{x} = 8.2$ minutes.

Questions/Tasks.

(C1) Identify whether this is a mean or proportion setting and define the parameter.

(C2) Write the null and alternative hypotheses for the claim.

(C3) Determine and justify the test tail type.

C1

This is a population mean setting. Let μ be the true mean cafeteria waiting time (in minutes) for students during lunch break. The benchmark is $\mu_0 = 9$. The sample information is $n = 64$ and $\bar{x} = 8.2$.

C2

$$H_0\colon \mu = 9, \qquad H_a\colon \mu < 9.$$

The null fixes the benchmark at 9 minutes, and the alternative reflects the improvement claim that average waiting time is lower.

C3

This is a left-tailed test because the directional claim is "less than 9 minutes." The alternative is expressed with $<$.

4 Problem 4 — Mean daily study time after a planning workshop

A Grade 11 coordinator wants to check whether the planning workshop changed average daily study time from the historical benchmark of 2.5 hours. A follow-up sample of $n = 50$ students gives $\bar{x} = 2.9$ hours.

Questions/Tasks.

(C1) Identify the parameter type and map symbols.

(C2) Write the null and alternative hypotheses.

(C3) Classify and justify the tail type.

C1

This is a population mean setting. Let μ be the true mean daily study time (in hours) for Grade 11 students after the workshop. The benchmark is $\mu_0 = 2.5$, and the sample information includes $n = 50$ and $\bar{x} = 2.9$.

C2

$$H_0 \colon \mu = 2.5, \qquad H_a \colon \mu \neq 2.5.$$

The null states no change from the benchmark. The alternative uses "different from" because both increases and decreases represent a change.

C3

This is a two-tailed test because the claim is about being different from 2.5 hours, not specifically higher or lower. The alternative uses $\neq$.

5 Problem 5 — Campus shuttle punctuality claim (direction check)

The transport office reports that at least 70% of morning shuttle trips arrive on time. A student auditor says the real punctuality may be lower than that benchmark. In a review of $n = 180$ trips, $x = 118$ were on time.

Questions/Tasks.

(C1) Identify the parameter and classify the setting.

(C2) Write a hypothesis pair that matches the auditor's directional claim.

(C3) Classify the tail type and justify why this is the correct direction.

C1

This is a population proportion setting. Let p be the true proportion of morning shuttle trips that arrive on time. The benchmark is $p_0 = 0.70$. The sample data are $n = 180$, $x = 118$, so $\hat{p} = 118/180 \approx 0.656$.

C2

$$H_0 \colon p = 0.70, \qquad H_a \colon p < 0.70.$$

This pair matches the auditor's concern that punctuality is lower than the official benchmark.

C3

Left-tailed test. The direction comes from the phrase "may be lower," which corresponds to $p < 0.70$. A right-tailed or two-tailed test would not represent the stated concern as directly.

6 Problem 6 — "I am almost there" travel-time message audit

A Grade 11 group studies travel messages before meetups. They define waiting time as minutes between a student texting "I am almost there" and actually arriving. They want to evaluate whether the true mean waiting time is greater than the practical benchmark of 7 minutes. Over several meetups, they collect $n = 55$ records and obtain $\bar{x} = 8.1$ minutes.

Questions/Tasks.

(C1) Identify the parameter and context setup.

(C2) Write null and alternative hypotheses in notation.

(C3) Classify and justify the test tail.

C1

This is a population mean setting. Let μ be the true mean waiting time (in minutes) between sending "I am almost there" and arriving at the meeting point. The benchmark is $\mu_0 = 7$, with sample information $n = 55$ and $\bar{x} = 8.1$.

C2

$$H_0\colon \mu = 7, \qquad H_a\colon \mu > 7.$$

The null keeps the practical benchmark at 7 minutes. The alternative formalizes the claim that the average delay is greater than that value.

C3

Right-tailed test, because the claim is specifically that the true mean waiting time is greater than 7 minutes. The alternative uses $>$, so the rejection region is in the upper tail.